

Braillix - Slice It Yourself (Anycubic Kobra Neo)

CURRENT RELEASE GATE - 2026-08-19: Do not manually slice production parts from this historical guide for tomorrow's run. Use only `printing/gcode_kobra_neo_checked/` and the exact contract in `PRINT_RELEASE_2026-08-19.md`. The checked profile is 0.16mm, 5 walls, 40% infill, 6/6 shells, 8mm brim, PETG 235/230C, bed 80C, supports off, and arc fitting off. A complete guide to producing your own G-code on your PC, so the lab just loads the file. Written 2026-07-30 for: **Anycubic Kobra Neo - 0.4mm nozzle - PETG - lab accepts G-code.**

Why your first print failed - the three real causes

What you saw	Actual cause
Supports welded in, couldn't pluck them out	Supports were switched on with PLA-tuned settings. PETG bonds across a gap that PLA snaps at. These parts need no supports at all.
Screw columns snapped	Not enough walls . On a dia 7.8mm column, walls are almost the whole part - infill barely matters.
Filament strings between travel moves	Stringing. Wet PETG + too-hot nozzle + wrong retraction. The Kobra Neo is direct drive , so most online retraction advice (5-6mm, meant for Bowden) is wrong and makes it worse.

None of these are your fault, and only one is the teacher's - they're the default settings.

PART 1 - Software

Install OrcaSlicer (recommended)

Download: <<https://github.com/SoftFever/OrcaSlicer/releases>> -> the .exe for Windows.

Why OrcaSlicer over Cura, for your situation:

- Anycubic Kobra profiles are **built in** - far fewer ways to get the machine setup wrong
- Has **built-in calibration tests** (retraction, temperature) that directly fix your stringing
- Per-object settings are much easier - which you need for the screw columns

Cura also works and your teacher may know it better. Cura equivalents for every setting are in **Appendix A** at the end.

First-run setup

- 1 Launch OrcaSlicer -> the setup wizard appears
- 2 **Printer** -> search Anycubic -> pick **Kobra Neo** (or **Kobra 2 Neo** - check the machine's

boot screen for which one). If it is genuinely absent, use **Appendix B** to add it manually.

- 1 **Filament** -> tick **Generic PETG**
- 2 Finish

Verify the machine settings (top-right printer dropdown -> the pencil/edit icon):

Setting	Value
Bed size	220 x 220 mm
Max height	250 mm
Nozzle diameter	0.4 mm
Extruder type	Direct Drive <- this one matters most

PART 2 - The settings that fix your problems

Switch OrcaSlicer to **Advanced** mode (top-right dropdown: Simple / Advanced / Expert -> choose **Advanced**), or you won't see half of these.

2.1 Quality tab

Setting	Value	Why
Layer height	0.2 mm	Good balance. 0.16 for the nicest surface, 0.28 if you're in a hurry.
First layer height	0.25 mm	Better bed adhesion
Wall loops	5	[!!] THE FIX FOR YOUR BROKEN SCREW COLUMNS. Default is 2. At 0.4mm each, 5 walls = 2mm of solid shell - on a dia 7.8mm boss that is nearly the entire column.
Top shell layers	5	Fixes the "exposed infill / rough top" you saw. Default 3 is too few.
Bottom shell layers	4	
Seam position	Aligned	Puts the layer-change blob in a consistent line instead of scattering it

2.2 Strength tab

Setting	Value	Why
Sparse infill density	25 %	Plenty. With 5 walls, infill is not carrying the load.
Sparse infill pattern	Gyroid	Strong in all directions, no sharp direction changes to shake the printer

2.3 Support tab - [!!] TURN IT ALL OFF

Setting	Value
Enable support	OFF (unchecked)

Nothing in these parts needs support, for two separate reasons:

1. The overhangs that could be solved by shape, are. Magnet pockets are **teardrop-shaped** (a 45deg peak instead of a round top) so they hold themselves up. Corner bosses have **gusset cones** widening toward the floor. Neither leaves anything to remove.

2. Everything else is just a normal bridge. Every hole in a wall is a 10-15mm span, and a well-cooled printer bridges **20-30mm** unaided:

Opening	Span	
Pogo slots (x2)	10mm	routine bridge
Wire exit	15mm	routine bridge
USB cutout (pod)	10mm	routine bridge
Nav holes (x3)	4.2mm	trivial

Note (v7.4): earlier versions modelled thin sacrificial bridges into these holes. They have been **removed** - they were redundant at these spans, and if you forgot to snap them out the pogo pins and wires could not pass through. **There is now nothing to remove from any hole.** The STL is exactly the finished part.

If you print with supports ON, the slicer fills the whole 60x60x50mm cavity with lattice that you cannot reach. That is exactly what happened to you.

2.4 Speed tab

Setting	Value	Why
Outer wall	30 mm/s	Slower = better surface
Inner wall	45 mm/s	
Sparse infill	60 mm/s	
Travel	150 mm/s	Fast travel = less time to ooze = less stringing
First layer	20 mm/s	Adhesion

2.5 Filament tab - [!!] THE STRINGING FIX

Setting	Value	Why
Nozzle temperature	235 degC (first layer 240)	If you still get strings, drop to 230 . Too hot is the #1 cause.
Bed temperature	80 degC	
Retraction length	1.0 mm	[!!] Kobra Neo is DIRECT DRIVE . Most guides say 5-6mm - that is for Bowden printers and will jam yours.
Retraction speed	35 mm/s	
Z-hop when retracting	0.2 mm	Stops the nozzle dragging through what it just printed
Wipe on retract	ON	

[!!] The thing that actually causes stringing: wet filament

PETG absorbs moisture from the air, and Patiala is humid. Wet filament strings no matter how perfect your settings are - the water boils out of the nozzle as steam.

Dry it before printing:

- Oven at **60-65 degC for 4-6 hours** (door slightly ajar). Do **not** exceed 70 degC or the spool warps.
- Or a food dehydrator at 65 degC
- Store it afterwards in a sealed box with silica gel

If the filament crackles or pops as it extrudes, it is wet. That is your problem, not settings.

PART 3 - Making JUST the screw columns solid

You asked exactly the right question: you want the screw columns denser without making the whole part 100 % infill. This is a **modifier**.

Honest answer first: with **Wall Loops = 5** the columns are already nearly solid, and that alone will probably fix the breakage. Try that before bothering with modifiers.

If you still want them fully solid:

- 1 Load `outer_box.stl`, **right-click** the model
- 2 **Add modifier -> Box**
- 3 A translucent box appears. In the right-hand **Object list**, select it.
- 4 Set its **size and position** so it encloses one corner boss:
 - Size: 12 x 12 x 40 mm
 - Position: X = 26, Y = 21, Z = 20 (that is one boss; the four are at X +/-26, Y +/-21)

- 1 With the modifier still selected, in the parameter panel below, click **+ Add setting ->**

Strength -> Sparse infill density -> set it to 100 %

- 1 **Repeat for the other three corners** (or select the modifier, Ctrl+C / Ctrl+V, and change the position)

Anything inside that box prints at 100 % infill; everything else stays at 25 %.

The same trick works for the mid-plate ledge or any other feature you want beefed up.

PART 4 - Orientation, part by part

[!!] `outer_box` **must be printed OPEN SIDE UP**. The rim sits on the bed, cavity facing the ceiling.

A previous version of our own notes said to flip it rim-down. **That was wrong and I've corrected it.** Nothing can bridge a 60mm cavity, and flipping it hangs the four dia 7.8 x 37mm bosses tip-first in mid-air - which is very likely why yours snapped.

Part	Orientation	Supports	Layer	Infill	Walls
outer_box	Open side UP, solid floor on bed	[X] OFF	0.16	40 %	5
base_plate	Flat, standoffs UP	[X] OFF	0.16	40 %	5
mid_plate	Flat, collar UP	[X] OFF	0.16	40 %	5
top_plate	HOLD pending top-interface coupon	[X] OFF	0.16	40 %	5
esp32_pod_shell	HOLD; open side UP, solid floor on bed	[X] OFF	0.16	40 %	5
esp32_pod_lid	HOLD; flat, ridges UP	[X] OFF	0.16	40 %	5
pogo_end_cap	HOLD; TPU 95A only, spool-specific profile required	[X] OFF	-	-	-

PART 5 - Producing the G-code

- 1 Load the STL: File -> Import -> Import STL, pick e.g. cad/stl/outer_box.stl
- 2 Orient it per Part 4. Rotate tool on the left toolbar; hold the mouse on a face and

use **Place on face** to drop a chosen face onto the bed.

- 1 Check the settings from Parts 2-3
- 2 Slice - the Slice plate button, bottom right
- 3 Inspect the preview. This is the step that would have saved your first print:

- Drag the **vertical layer slider** on the right up and down
- Look for **blue/teal support material** - if you see any, supports are still on. Turn them off.
- Check the bridges over the pogo slots and wire exit actually appear as thin flat spans
- Confirm the corner bosses look solid, not hollow

- 1 Export G-code -> save to an SD card as e.g. outer_box.gcode
- 2 Hand the SD card to the lab. They just load and print.

Print ONE part first - pogo_end_cap or mid_plate. Small, fast, and it tells you whether your settings are right before you commit 3 hours to the box.

PART 6 - After the print

Job	How
Clean the screw holes	The M2.5 pilots print ~0.2-0.3mm undersize. Run a 2.5mm drill bit through by hand - do not power-drill, you'll melt it.
Magnet pockets	Should be clear (teardrop shape). If tight, a 8mm bit turned by hand.
Stringing wisps	Snip with flush cutters, or wave a heat gun / lighter quickly past them (do not hold it there)
Layer lines	400 -> 800 grit wet sandpaper if you care cosmetically

PART 7 - How to close the Brain Pod

The pod is two printed parts: **shell** (the open box) and **lid**.

<pre> +-----+ POD LID o o +-----+ +-----+ POD SHELL [ESP32 sits on header strips] ? ? +-----+ </pre>	<pre> <- 3 ridges on top (tactile ID) <- 2 screw holes at X = +/-27.5 the lid's front/back SKIRT drops over the shell and self-locates it <- 2 screw BOSSES, also at X = +/-27.5 </pre>
--	--

Steps:

- 1 ESP32 plugged into its header strips inside the shell, wires routed
- 2 Lower the lid straight down - the **skirt on the front and back edges** slides over the shell walls and centres it. The left/right ends stay flush (those are the docking faces).
- 1 The two holes line up over the two bosses
- 2 Drive **2 x M2 x 8mm self-tapping screws** down through the lid into the bosses

Screw length matters: the lid is 4mm thick and the boss is 8mm deep. An **M2x8** leaves 4mm of thread biting into the boss. An M2x6 only gives 2mm and will strip out - do not use it.

No glue. The pod is meant to open so you can swap the ESP32.

Appendix A - Cura equivalents

OrcaSlicer	Cura
Wall loops	Wall Line Count
Sparse infill density	Infill Density
Enable support	Generate Support
Retraction length	Retraction Distance
Top shell layers	Top Layers
Modifier box	Per Model Settings -> *Modify settings for overlaps*
Slice plate	Slice

In Cura you must also set the cog icon -> **Setting Visibility** -> **Expert** or most of these stay hidden.

Appendix B - Manual Kobra Neo profile (only if it's not built in)

Add Printer -> Custom / Add local printer:

Build volume	220 x 220 x 250 mm
Nozzle	0.4 mm
Filament	1.75 mm
Extruder	Direct drive
Bed shape	Rectangular, origin front-left
Firmware / G-code flavour	Marlin
Start G-code	leave the default Marlin one

The five-line version

- 1 **Supports OFF** - every overhang is already handled in the CAD
- 2 **Wall loops = 5** - this is what fixes the broken screw columns
- 3 **Retraction 1.0mm** - Kobra Neo is direct drive, not Bowden
- 4 **Dry the PETG** - 60-65 degC for 4-6 h; wet filament strings regardless of settings
- 5 **outer_box prints OPEN SIDE UP** - never rim-down